



Full length article

IoT-FAR: A multi-sensor fusion approach for IoT-based firefighting activity recognition

Xiaoqing Chai^a, Boon Giin Lee^{a,*}, Chenhong Hu^a, Matthew Pike^a, David Chieng^b, Renjie Wu^c, Wan-Young Chung^d^a Nottingham Ningbo China Beacons of Excellence Research and Innovation Institute, School of Computer Science, University of Nottingham Ningbo China, Ningbo 315100, China^b Department of Electrical and Electronic Engineering, University of Nottingham Ningbo China, Ningbo 315100, China^c Department of Civil Engineering, The University of Hong Kong, 999077, Hong Kong, China^d Department of Electronic Engineering, Pukyong National University, Busan 48513, Republic of Korea

ARTICLE INFO

Dataset link: <https://github.com/HCI-Laboratory/Smart-Firefighting/tree/main/Firefighter-Activity-Recognition>

Keywords:

Human activity recognition

Sensor fusion

Machine learning

Wearable computing

Internet of things

ABSTRACT

Inadequate training poses a significant risk of injury among young firefighters. Although Human Activity Recognition (HAR) algorithms have shown potential in monitoring and evaluating performance, most existing studies focus on daily activities and have difficulty distinguishing complex firefighting tasks. This study introduces the Internet of things (IoT)-based wearable firefighting activity recognition (IoT-FAR) system which employs a multi-modal sensor fusion approach to achieve comprehensive activity recognition during firefighting training. The IoT-FAR comprises five wearable body sensor nodes and a coordinator node. This study explores the significance of features extracted from the surface electromyography, heart rate, and inertial measurement units in firefighting training activity recognition. A hybrid machine learning (HML)-based network is proposed, which integrates three models: one trained with all features (MA), another with upper body features (MU), and a third with lower body features (ML). The proposed HML-SVM-RBF1-RF2 network achieves superior performance, with a mean recall of 93.94%, mean precision of 90.94%, and a mean accuracy rate of 98.29%. Additionally, the study introduces the specialized firefighting training associated activities (SFTAA) dataset, which includes endurance training activities involving self-contained breathing apparatus (SCBA) conducted by eighteen firefighters. This dataset represents preliminary work towards building a comprehensive dataset covering various events and scenarios for tracking firefighter activities. The IoT-FAR system also demonstrates the potential use of misclassified activities as evaluation metrics for assessing firefighter training performance.

1. Introduction

Firefighting is a demanding and dangerous profession requiring significant physical and mental fitness, along with professional competence for effective rescues. According to the annual firefighting report in China [1], there were 111 firefighter casualties between 2010 and 2020, with an average age of 27 years. Inadequate training and practical experience, leading to poor decision-making, are major factors contributing to injuries [2]. Specialized firefighting training is crucial for enhancing rescue skills, physical fitness, and tactical abilities. However, the scientific severity of the training approach profoundly influences firefighters' physical aptitude and combat effectiveness.

Research has highlighted a lack of innovative physical fitness training in many fire brigades, with a reliance on traditional military

methods, which poses challenges in motivating firefighters [3]. Additionally, traditional training has been shown to contribute to firefighter fatigue [4]. For instance, self-contained breathing apparatus (SCBA) endurance training is vital to acclimate firefighters to tasks at the fire-ground and improve their physical fitness and resilience. The training assessment criterion requires performing at least 20 activities from full capacity (28–29 mPa) to the warning state (less than 8 mPa) of oxygen level. However, individuals with poor physical fitness may use non-standard methods to conserve energy, leading to training inefficiency and safety concerns during actual firefighting.

Over the past decade, there has been significant interest in human activity recognition (HAR) within the broader research community.

* Corresponding author.

E-mail addresses: xiaoqing.chai@nottingham.edu.cn (X. Chai), boon-giin.lee@nottingham.edu.cn (B.G. Lee), scych6@nottingham.edu.cn (C. Hu), matthew.pike@nottingham.edu.cn (M. Pike), david.chieng@nottingham.edu.cn (D. Chieng), rjwu@hku.hk (R. Wu), wychung@pknu.ac.kr (W.-Y. Chung).<https://doi.org/10.1016/j.inffus.2024.102650>

Received 10 July 2024; Received in revised form 9 August 2024; Accepted 20 August 2024

Available online 25 August 2024

1566-2535/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

HAR aims to distinguish human behavior by analyzing data from various sensors [5]. This field has become crucial across applications such as healthcare, surveillance, sports tracking, and robotics [6]. Nevertheless, most prior HAR studies have focused on classifying routine activities of daily living (ADLs), such as walking, running, and jogging [7]. Improving classification performance in more complex activities remains a notable challenge and is a subject of ongoing research. HAR has the potential to enhance firefighter training in several ways. It can provide real-time feedback to trainers, allowing them to monitor firefighters' behavior during training, thereby improving safety and assessment. Additionally, it enables remote monitoring and assessment of performance by supervisors or trainers. Furthermore, using a data-driven approach, HAR can aggregate firefighter performance data over time, enabling customized training based on individual analyses from previous sessions.

However, applying HAR in firefighting training presents unique challenges compared to usual ADLs. These challenges arise from the inherently complex nature of firefighting activities, which involve a broader range of movements and faster speeds. For instance, unlike conventional stair climbing, firefighters may need to carry two trays of hoses in both hands simultaneously to simulate real firefighting scenarios. While recent studies emphasize deep learning (DL)-based approaches for improved recognition accuracy, it is essential to acknowledge that these methods come with increased computational costs and a risk of model overfitting [8].

This study aims to evaluate the performance of HAR models in classifying firefighting activities during SCBA endurance training, utilizing wearable sensors positioned at multiple locations on the body. The objective is to propose an IoT-based wearable firefighting activity recognition system (IoT-FAR) capable of accurately classifying firefighting activities, enabling timely monitoring of training performance by trainers. The contributions of this paper are outlined as follows:

- An Internet-of-Firefighting (IoF)-based wearable Firefighting Activity Recognition (FAR) architecture is proposed. This architecture integrates a wireless body sensor network (WBSN) with sensors placed on the arms and legs, and a coordinator attached to the self-contained breathing apparatus (SCBA). This configuration facilitates the collection of motion and physiological data from firefighters.
- A novel hybrid machine learning (HML)-based network is developed for classifying complex firefighting activities during SCBA endurance training. The study highlights the importance of sensor placement by analyzing its impact on the FAR classification performance.
- A publicly available dataset, the specialized firefighting training associated activities (SFTAA) dataset, is built and used for model testing. This dataset includes sensor data from eighteen firefighters engaged in SCBA endurance training activities, including motion, heart rate (HR), surface electromyography (sEMG), and environmental data. The training activities are categorized into upper and lower body parts.

These contributions collectively form a comprehensive framework for advancing the understanding and implementation of IoT-FAR technology in specialized firefighting training.

The rest of this paper are structured as follows: Section 2 provides a review of related FAR and HAR studies. Section 3 presents an overview of the IoT-FAR system architecture, details the data collection process, and introduces the proposed HML-based network. Section 4 outlines the experimental results and evaluates the performance of the implemented networks. Concluding remarks are presented in Section 5, summarizing key insights and outlining potential directions for future research.

2. Related work

2.1. Sensor-based HAR

HAR is broadly categorized into vision-based and sensor-based methods [9]. The vision-based approach uses images and videos from optical sensors to identify activities [10]. Although this method is relatively easy to implement and understand, it has significant drawbacks, such as privacy issues, the need for pre-installed hardware in the environment, and the requirement for unobstructed views [11]. Consequently, there is a growing preference among researchers for the sensor-based approach, which utilizes various sensors to measure user motion, location, environmental information, and human-object interaction [12]. The sensor-based method is particularly suitable for firefighting scenarios with low visibility due to heavy smoke and high temperatures.

Many researchers have developed HAR models to classify ADLs using smartphones and smartwatches equipped with inertial measurement units (IMUs) to analyze user motion data [13,14]. These HAR systems are particularly beneficial in healthcare settings as they do not require additional devices. For instance, Ran et al. [15] proposed a DL-based HAR system that classifies seven ADLs using 9 degree-of-freedom (9-DOF) IMU data from four smartphones placed in different locations. Similarly, Yande et al. [16] introduced a smartwatch-based HAR method to classify five activities, including stay, walk, run, up-stairs, and downstairs. They also emphasized that the performance of smartphone-based HAR is significantly influenced by the smartphone's placement compared to smartwatch-based HAR. However, HAR using smartphones and smartwatches may struggle to distinguish complex ADLs, making them less suitable for firefighting training.

In contrast, numerous studies have proposed sensor fusion methods to enhance accuracy by directly integrating data from multiple sensors or employing feature or decision-level fusion techniques [17]. Sen et al. [18] utilized multiple micro-electro-mechanical systems (MEMS) IMU sensors, or combining them with other sensors placed at various body locations, to develop a comprehensive HAR system, similar to other studies [19–21]. Meanwhile, some studies have introduced gait pattern recognition methods that distinguish between walking and running, along with path trajectories, based on the fusion of IMUs positioned at the ankles or feet [22–24]. Conversely, Ran et al. [25] proposed a positioning method for locating a group of mobile users through the fusion of IMU and radio measurements. Tao et al. [12] proposed a human-object interaction method for HAR, integrating three accelerometers on the hands and waist, along with two radio-frequency identification (RFID) wristband readers on each hand. The RFID readers detect RFID tags attached to household objects, such as cups and spoons, enhancing classification performance.

Additionally, Bangaru et al. [26] investigated HAR using a Myo armband in conjunction with a 9-DOF IMU and surface electromyography (sEMG) sensors worn on the forearm. They demonstrated that the combined model with sEMG and IMU achieves higher accuracy than individual sensor models through performance analysis and long-term reliability tests. Vijayvargiya et al. [27] proposed a HAR method by measuring and analyzing lower limb sEMG signals using explainable artificial intelligence (XAI). Despite these advancements, the practical performance of sEMG in activity recognition remained under discussion, as most prior experiments collected data in static positions, which facilitated consistent muscle contractions throughout the trials [28].

Moreover, Pei et al. [29] introduced a novel HAR approach utilizing virtual IMU data of a virtual human from the skinned multi-person linear (SMPL) model in a virtual environment. The synthetic data was transferred to real IMU data using transfer learning. Another innovative HAR approach involved wearable suspenders integrated with strain sensors, as discussed in [30]. However, while these studies focus on predicting simple ADLs, the exploration of the number and placement of sensors for complex activity recognition remains limited.

2.2. HAR classification

ML and DL approaches are mainly adopted in the classification methods used in HAR. Recent studies have focused on combining DL models with multi-modal network structures, particularly the Convolutional Neural Network with Long Short-Term Memory (CNN-LSTM) model [31,32]. CNNs automatically extract features, while LSTMs, a type of Recurrent Neural Network (RNN), are well-suited for time-series data. To increase performance, attention mechanisms and residual blocks have been applied, leading to models such as 1DCNN-ResBLSTM-Attention [33], multilevel residual network with attention (Multi-ResAtt) [34], Transformer with bi-directional GRU (TRANS-BiGRU) [5], multi-head attention and graph neural networks (MhaGNN) [35], and channel attention-based deep residual network (ResNet-SE) [36].

DL approaches generally achieve greater performance in HAR, but their high computational cost and the optimization of hyperparameters makes deployment on embedded systems or mobile devices challenging [18]. To address this, studies like [37,38] have proposed lightweight CNNs to reduce computational costs. The rapid development of system on a chip (SoC) technology has enabled the deployment of ML models on edge devices, leading to the evaluation of real-time HAR systems using various ML methods. Commonly used methods include k-nearest neighbor (KNN) [39], decision tree [40], support vector machine (SVM) [41,42], and random forest (RF) [43].

Researchers have also shifted their focus to classifying more complex activities than basic ADLs, termed complex HAR (CHAR). Multi-modal approaches combined with various ML or DL methods are emphasized to achieve competitive performance in CHAR. Hierarchical methods have been proposed in studies such as [44,45], which first distinguish coarse-level categories and then fine-grained subcategories. For instance, Qi et al. [46] proposed a two-layer approach where the first layer of SVM distinguishes between free and non-free weight activities, and the second layer employs a neural network to classify detailed aerobic and sedentary activities. The hidden Markov model (HMM) is applied for further refinement of free weight activities. Alternatively, Jeya et al. [47] introduced X-CHAR, a concept-based CHAR using DL models, which models the activity as a sequence of concepts, resulting in a more understandable and explainable DL model than previous approaches.

While previous studies have introduced diverse DL approaches, they often exhibit data-intensive requirements and encounter significant computational costs, making them unreliable and unsuitable for practical application in real firefighting scenarios.

2.3. HAR applications

Most existing HAR studies focus on applications in ambient assisted living (AAL) systems, primarily tracking ADLs among residents, particularly the elderly [43]. HAR has proven effective in various fields. For example, in the field of sports and fitness, Nguyen et al. [48] proposed a smartphone-based HAR system to ensure cyclists' safety by recognizing activities such as standing, sitting, walking, cycling, and falling. In a different context, Mastakouris et al. [49] proposed an LSTM-based HAR model to recognize five factory worker activities, although it required workers to attach a smartphone to their right wrist. Mekruksavanich et al. [50] achieved recognizing sixteen construction worker activities utilizing three 9-DOF IMUs placed at the hip, upper arm, and the rear of the shoulder.

Despite extensive literature on HAR, only a few studies specifically address HAR systems in firefighting. Geng et al. [42] proposed an innovative on-body radio frequency features-based HAR system that uses SVM to monitor firefighters' physiological conditions. A distinguished AIA'15 data mining competition [51], "Tagging Firefighter Activities at a Fire Scene", which was organized at the Knowledge Pit platform, aimed to classify common firefighter activities into binary categories

(posture and action) and twenty-one subcategories, highlighting HAR's significance in firefighting activity classification. However, the competition relied on multi-positional sensors with wired connections, which could affect mobility, and the dataset is no longer available.

This study uses multi-positional wearable sensors equipped with sEMG and IMU sensors for firefighting training classification. This method replaces vision-based and ambient-based approaches that depend on pre-installed equipment, impractical in firefighting's challenging environment. sEMG sensors measure muscle activities during physical firefighting tasks, while HR data are collected to examine the correlation between physical tasks and HR changes. Inspired by the hierarchical method discussed earlier, this study introduces a novel HML-based network to improve classification performance and enable real-time monitoring. Establishing a reliable real-time HAR system for classifying firefighting training activities represents a significant advancement in efficiency and introduces an evaluative approach to firefighter training.

3. Methods

3.1. Hardware design

Previous research [52] introduced the IoF, an IoT-based firefighting application using Wi-Fi and LoRaWAN technologies for data transmission. The study revealed that the heterogeneous IoF network improved data transmission throughput and range across various firefighting scenarios but faced increased Wi-Fi power consumption. To address this issue, the current study replaces Wi-Fi with Bluetooth Low Energy (BLE) for data transmission using a WBSN.

Fig. 1 shows the system architecture of the proposed IoT-FAR system integrated with the updated IoF framework. Table 1 details the system components, including five sensor nodes (SNs) and a custom-designed body node coordinator (BNC). The five SNs consist of four customized sEMG and IMU sensor bands (EMG-IMU), along with a HR armband [53] for HR monitoring (HR-SN). These EMG-IMU bands are positioned on different parts of the firefighter's body, covering the lower left forearm (LA), lower right forearm (RA), left shank (LS), and right shank (RS). The HR armband is placed on the upper left forearm (UA). Additionally, the BNC contains an IMU sensor for the FAR, which is placed on the SCBA and denoted as B, in addition to LA, RA, LS, and RS. The components of an EMG-IMU band include:

- MUP6050 module [54]: This 6-DoF IMU includes an accelerometer and a gyroscope to measure the firefighter's acceleration and angular velocity for capturing motion data.
- sEMG module [55]: Records sEMG signals from the brachioradialis and gastrocnemius muscles, for distinguishing tasks such as lifting hoses and breaking down doors.
- BLE 4.2 module (JDY-18) [56]: Enables seamless communication with the BNC at a sampling rate of 100 Hz.

The armband on the UA measures the firefighter's photoplethysmography (PPG)-based HR, serving as a physiological indicator of training intensity. The armband transmits data to the BNC at a sampling rate of 1 Hz. The BNC is the central component of the IoT-FAR system, housing the FAR model deployment. To ensure minimal interference with the firefighter's mobility, the BNC is composed of three modules and is affixed to the SCBA:

- RPi-Embedded Module: Configured with a Raspberry Pi (RPi) 4B+ with 4 GB RAM [57], this module collects data from the five SNs via BLE connection and executes the FAR model.
- LoRaWAN Module: Features a LoRa wireless area network (WAN) module with a LoRaWAN gateway [52], establishing the LoRa connection between the indoor (WBSN) and outdoor (terminal).
- Multi-Sensor Board: This customized extension board, attached to the RPi, integrates the TinyPICO micro-processing unit [58], a BNO055 IMU (capturing motion data of the body torso) [56], and the BME688 environmental sensor (collecting humidity, temperature, pressure, and volatile organic compounds data) [59].

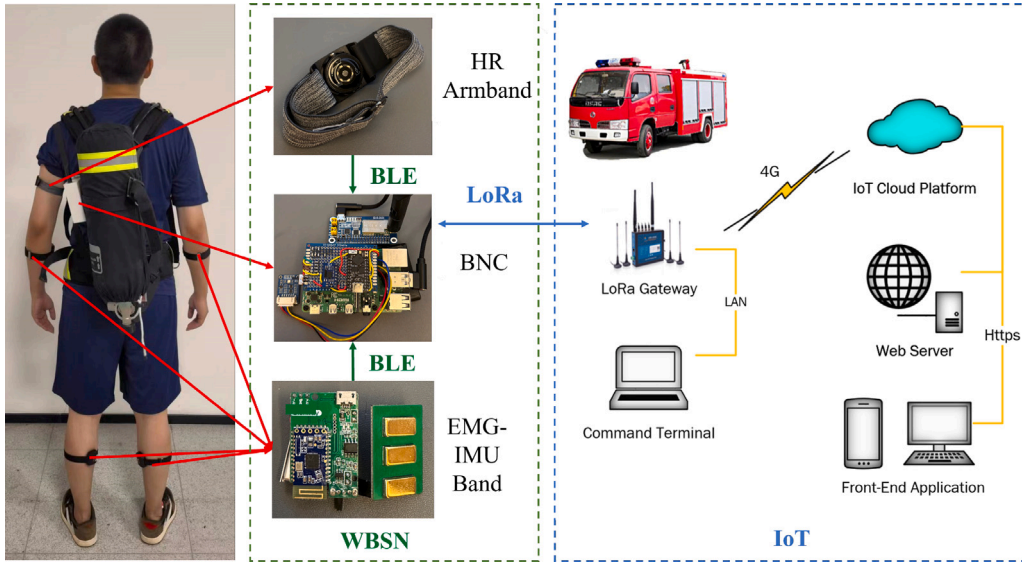


Fig. 1. Overview of the proposed IoT-FAR system, including the FAR and the IoF network framework.

Table 1
Hardware specifications.

Module	Component	Specification
EMG-IMU	MPU6050	6-DOF IMU; Sampling rate: 100 Hz
	sEMG	Three dry electrodes (DRY+, DRY-, and GND)
	JDY-18	BLE 4.2
BNC	LoRaWAN	ASR6601 SoC; Frequency range: 398–525 MHz; Data rate: 0.814 Kbp
	RPi	1.8 GHz Quad core Arm Cortex-A72; 4 GB RAM; 2.4 GHz Wi-Fi & BLE 5.0
	BNO055	9-DOF IMU
	BME688	Measures humidity, temperature, pressure, and volatile organic compounds
	TinyPICO	240 MHz Dual core Xtensa LX6; 4 MB RAM; 2.4 GHz Wi-Fi & BLE 4.2

3.2. Building of SFTAA dataset

Eighteen volunteer firefighters from the Huzhou Fire Brigade in Zhejiang province, China, participated in this study, wearing full firefighting equipment during SCBA endurance training activities. The participants had a mean age of $22 \pm$ years (ranging from 19 to 28 years), a mean height of 174 ± 10 cm (ranging from 166 to 184 cm), and a mean weight of 75 ± 15 kg (ranging from 64 to 90 kg). The study adhered to ethical guidelines and received approval from the university's research ethics committee, complying with the university's code of research conduct and ethics.

The SCBA endurance training comprised seven specialized activities (A1 to A7) and one stationary activity (A8), as illustrated in Fig. 2. The details of these activities are outlined in Table 2. The experiments were conducted in controlled fire scenarios without actual fire and smoke, as shown in Fig. 3. On the left side of the large empty field, a regular vehicle tire was placed to simulate an obstacle that needed to be broken (see A1). On the right side, a 60 kg dummy was positioned to represent a victim awaiting rescue (see A2). A fire engine equipped with a water supply and hose was used to simulate an outdoor firefighting activity

(see A3). In the center of the field, a fire training tower was used for firefighters to perform various activities (see A4, A5, and A6).

Each firefighter was instructed to complete the entire SCBA endurance training with a full oxygen level capacity of 28 mPa. Completing activities A1 to A6 was considered one cycle, which was to be performed continuously without breaks. If the SCBA oxygen level remained above 8 mPa (not reaching the warning level), the firefighter was instructed to begin a new cycle. Nevertheless, upon receiving a warning alert from the SCBA, the firefighter was required to stop training and proceed to A8 and resting for approximately two to three minutes before removing the firefighting equipment.

In accordance with the training evaluation standard, each firefighter was required to complete at least 20 activities, equivalent to a minimum of three cycles. The experiment duration for each firefighter varied based on individual physical oxygen consumption ability. The results indicated that only three out of the eighteen firefighters met the training evaluation standard, highlighting the challenge of the SCBA endurance training. A firefighter instructor was present throughout the training sessions to ensure safety of firefighters. Note that some firefighters exhibited signs of fatigue or lower physical fitness and chose to stand during activity A8 (instead of sit or squat), which was acceptable per the instructor's guidance. Data cleaning and labeling was conducted by the two main researchers of this study and validated by the firefighter training instructors.

3.3. Data pre-processing

During the preliminary analysis, it was observed that trained firefighters consistently exhibited high HRs with low variability in HR data which resulting in low significance for FAR prediction based on firefighter conditions. Additionally, since SCBA endurance training was conducted in a controlled environment lacking realistic firefighting scenarios with actual fire and smoke, environmental and HR data were excluded from the FAR analysis in this study.

The time-series raw data from the IMU at time step t for the sensor node S can be represented as A_t^S , G_t^S , and E_t^S , defined as:

$$A_t^S = [Ax_t^S, Ay_t^S, Az_t^S] \in \mathbb{R}^3, \quad (1)$$

$$G_t^S = [Gx_t^S, Gy_t^S, Gz_t^S] \in \mathbb{R}^3 \quad (2)$$

where Ax_t^S , Ay_t^S , and Az_t^S represent the acceleration forces (m/s^2) along the x -, y -, and z -axis (including gravity), respectively. Similarly,



Fig. 2. Illustration of the eight activities in the SCBA endurance training: (a) A1, (b) A2, (c) A3, (d) A4, (e) A5, (f) A6, (g) A7 and (h) A8.



Fig. 3. Experiment setup of the SCBA endurance training.

Table 2
Details of firefighting SCBA endurance training activities.

Activity	Description
A1	Use a hammer to hit a tire, moving it 5 m to simulate a forceful break-in action.
A2	Drag a dummy for 20 m to simulate the rescue of a trapped person.
A3	Move the nozzle with a water-supplied hose for 20 m to simulate an outdoor firefighting task.
A4	Climb from the ground floor to the fourth floor while carrying two trays of hoses to simulate firefighting tasks in a high-rise building.
A5	Drag the hose from the ground floor to the fourth floor via window to simulate external replenishment tasks in indoor firefighting.
A6	Descend from the fourth floor to the ground floor while carrying two trays of hoses to simulate firefighting tasks on different floors.
A7	Perform usual walking movement.
A8	Engage in stationary actions, including squatting or sitting during or after training to simulate waiting for peer rescue in dangerous situations.

Gx_t^S , Gy_t^S , and Gz_t^S represent the rotational rates (rad/s) around the x -, y -, and z -axis, respectively, and $S \in \{B, LA, RA, LS, RS\}$. With the inclusion of sEMG data on four nodes placed on the limbs, denoted by E_t^S , the sequence of raw data for S can be described as:

$$R^S = \begin{cases} ([A_t^S, G_t^S])_{t=1}^T \in \mathbb{R}^{6 \times T}, S \in \{B\} \\ ([A_t^S, G_t^S, E_t^S])_{t=1}^T \in \mathbb{R}^{7 \times T}, S \in \{LA, RA, LS, RS\} \end{cases} \quad (3)$$

where T is the sequence length, and sEMG data is not measured for sensor node B (placed on the body trunk). The 6-DOF IMU data was then transformed into the signal magnitude vector (SMV) [60] and the angular axis of acceleration vector (AAV) [61]. The SMV is defined by

Eqs. (4) to (5), and the AAV is defined by Eq. (6):

$$SMV a_t^S = \sqrt{(Ax_t^S)^2 + (Ay_t^S)^2 + (Az_t^S)^2} \quad (4)$$

$$SMV g_t^S = \sqrt{(Gx_t^S)^2 + (Gy_t^S)^2 + (Gz_t^S)^2} \quad (5)$$

$$AAV_t^S = \cos^{-1}(A_t^S / Av_t^S) \quad (6)$$

where Av_t^S denotes the three-axis acceleration force vector, and the processed data P_t^S is given by:

$$P^S =$$

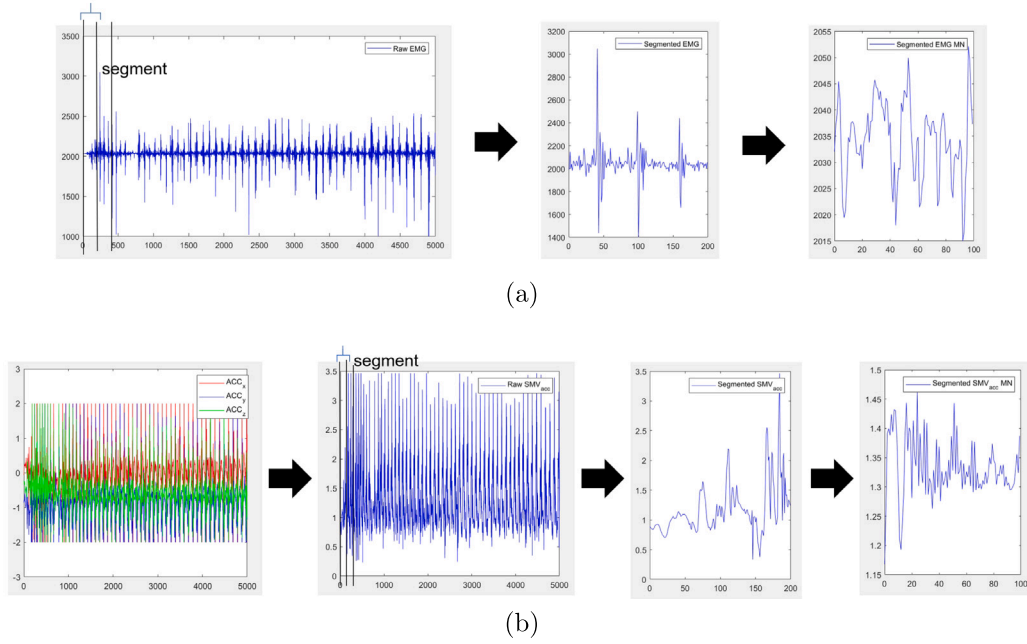


Fig. 4. Data pre-processing stages for the LA node, illustrating (a) EMG and (b) ACC data segmentation and feature extraction.

$$\begin{cases} ([SMV a_t^S, SMV g_t^S, AAV_t^S])_{t=1}^T \in \mathbb{R}^{5 \times T}, S \in \{B\} \\ ([SMV a_t^S, SMV g_t^S, AAV_t^S, E_t^S])_{t=1}^T \in \mathbb{R}^{6 \times T}, S \in \{LA, RA, LS, RS\} \end{cases} \quad (7)$$

The study evaluated various window sizes and selected a 2 s window (W) with a 1.5 s overlap (V) for feature extraction segmentation. In addition to the four features used in previous work [62], this study included a total of 31 features, comprising 25 time-domain features and 6 frequency-domain features. The SHapley Additive exPlanations (SHAP) method [63] was then employed to assess the importance of features associated with IMU and EMG data. Table 3 illustrates the final selected features $F_{a,n}^S$, denoted as:

$$F_{a,n}^S = f_a(P_n^S) = f_a(P_{(n-1) \times V + 1 : W + (n-1) \times V}^S) : n \in N \quad (8)$$

where a is the index of a feature, n is the segments of features, N represents the set of segments, and $f_a(P_n^S)$ represents the function to calculate the a th feature at n th segment.

In total, 654 features were extracted for each window segment across five sensor nodes (excluding one sEMG sensor node, as shown in Fig. 1). Fig. 4 shows the data pre-processing steps for EMG and accelerometer data on the LA node, including data segmentation and feature extraction.

3.4. Proposed HML-based network

This study introduces an HML-based network tailored to the unique sensor placements on firefighters, as depicted in Fig. 5, comprising three distinct models: one integrating all selected features (MA), another focusing solely on upper body (UB) features (MU), and a third emphasizing lower body (LB) features (ML). The features for UB and LB are denoted as $FUB_{b,n}$ and $FLB_{c,n}$, respectively, and are represented by:

$$FA_{a,n} = \sum F_{a,n}^S = [FUB_{b,n}, FLB_{c,n}] \quad (9)$$

where b and c are the indices of the features for UB and LB, respectively.

The UB activities are further categorized based on muscle force usage into instantaneous force (UB1), continuous force (UB2), and irrelevant activities (UB3). Similarly, LB activities are classified based

on leg movement patterns, including climbing (LB1), walking (LB2), and stationary positions (LB3).

Table 4 outlines the UB and LB sub-activities corresponding to main activities A1 to A8, alongside the number of segmented data instances obtained post data pre-processing and validation via recorded videos, totaling 26,947 segmented data instances. Note that the activity durations vary among firefighters; for instance, activities involving climbing (A4) or descending stairs (A6) span longer durations, resulting in increased segmented data instances, as shown in Table 4. Despite this variance, data collection complied with an individual-centric approach, with all firefighters undergoing identical SCBA endurance training. While this approach may lead to slight class imbalances across the eight activities, it does not significantly impact the model's training and testing phases.

The final step involves a weighted combination to determine the probability of each activity by aggregating predictions from the three models using predefined weights, formulated in Eq. (10):

$$P(F_i) = P(A_i) + P(U_{i|j}) \times \bar{U}_{i|j} + P(L_{i|k}) \times \bar{L}_{i|k} \quad (10)$$

where $i \in \{A1 \dots A8\}$, $j \in \{U1, U2, U3\}$, $k \in \{L1, L2, L3\}$, $P(A)$, $P(U)$, and $P(L)$ denote the predicted probabilities of the MA, MU, and ML models, respectively. $\bar{U}$ and $\bar{L}$ represent the mean accuracies of the MU and ML models, respectively. Here, $i|j$ and $i|k$ signify the corresponding combination of UB and LB sub-activities. For instance, for A1:

$$P(F_{A1}) = P(A_{A1}) + P(U_{A1|U1}) \times \bar{U}_{A1|U1} + P(L_{A1|L3}) \times \bar{L}_{A1|L3} \quad (11)$$

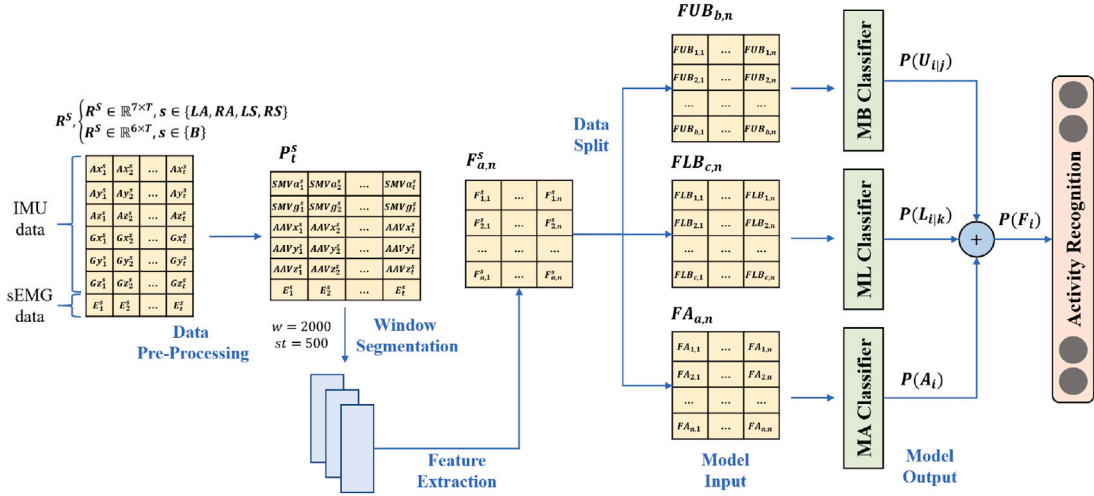
3.5. Classifier selection

To assess the effectiveness of the proposed Hybrid Machine Learning (HML) network, this study initially focused on selecting the most suitable MA classifiers (see Fig. 5), taking into account their accuracy and processing time, both of which are essential for embedded real-time activity classification. Four classifiers were selected for evaluation: KNN [39], DT [40], SVM [41], and RF [39]. The hyperparameters of these classifiers were examined and adjusted to identify the optimal configurations for the proposed network. These configurations varied across classifiers, with the optimal hyperparameters for the MA, MU, and ML models detailed in Table 5. For instance, models like KNN and DT presented different configuration of hyperparameters for MA,

Table 3

Selected features including 25 time-domain features and 6 frequency domain features.

Domain	Feature	SMV_{ACC}	SMV_{GYR}	AAV_x	AAV_y	AAV_z	EMG
Time	Mean (MN)	✓	✓	✓	✓	✓	✓
	Standard Deviation (STD)	✓	✓	✓	✓	✓	✓
	Range (RG)	✓	✓	✓	✓	✓	✓
	Mean Absolute Deviation (MNAD)	✓	✓	✓	✓	✓	✓
	Median Absolute Deviation (MDAD)	✓	✓	✓	✓	✓	✓
	Mean Absolute Value (MAV)	✓	✓	✓	✓	✓	✓
	Simple Square Integral (SSI)	✓	✓	✓	✓	✓	✓
	Root Mean Square (RMS)	✓	✓	✓	✓	✓	✓
	Average Amplitude Change (AAC)	✓	✓	✓	✓	✓	✓
	L2norm (L2N)	✓	✓	✓	✓	✓	✓
	L-Scale (LSC)	✓	✓	✓	✓	✓	✓
	Difference Absolute Standard Deviation (DASTD)	✓	✓	✓	✓	✓	✓
	Interquartile Range (IQR)	✓	✓	✓	✓	✓	✓
	Skewness (SKEW)	✓	✓	✓	✓	✓	✓
	Signal Magnitude Area (SMA)	✓	✓	✓	✓	✓	✓
	Kurtosis (KURT)	✓	✓	✓	✓	✓	✓
	Autocorrelation (CORR)	✓	✓	✓	✓	✓	✓
	Auto-Regressive Coefficient (ARC)	✓					✓
	Logarithm Base 10 (LOG)						✓
	Zero Crossing (ZC)						✓
	Sequential Segments Change (SSC)						✓
	Myopulse Percentage Rate (MYOP)						✓
	Willison Amplitude (WAMP)						✓
	Hjorth Mobility (HJM)	✓		✓	✓	✓	✓
	Hjorth Complexity (HJC)	✓		✓	✓	✓	✓
Frequency	Mean Frequency (MNF)	✓	✓				✓
	Median Frequency (MDF)	✓	✓				✓
	Spectral Peak Height (SPH)	✓	✓				✓
	Peak Frequency (PF)	✓	✓				✓
	Total Power (TP)	✓	✓				✓
	Variance of Central Frequency (VCF)	✓	✓				✓
Total features per SN		26	23	19	19	19	31
Total SN		5	5	5	5	5	4
Total features		130	115	95	95	95	124

**Fig. 5.** Overview of the proposed HML-based network flowchart.

MU, and ML, while others upheld consistent configuration of hyperparameters across these models. Additionally, two neural networks were investigated: the CNN-LSTM, a widely used network in HAR [9], and Multi-ResAtt [34], known for its high performance in multi-sensor fusion. It is also important to note that the architectures of CNN-LSTM and Multi-ResAtt networks were specifically redesign to reflect firefighting task characteristics rather than generic human activities.

For model training and testing, a high-performance desktop PC equipped with an Intel Core i7-13700K 3.4 GHz CPU and NVIDIA RTX 4090 24 GB RAM GPU was utilized. Cross-validation was utilized to assess individual model performances.

4. Results and discussion

This study evaluates the performance of the HML-based network using four metrics, including macro-average recall (RC), macro-average precision (PR), macro-average F1 score ($F1$), and mean accuracy rate (AC). These metrics are preferred for their practical applicability in real-world scenarios, which are defined as:

$$AC = (1/N) \sum_{i=1}^N (TP_i + TN_i) / (TP_i + TN_i + FP_i + FN_i) \quad (12)$$

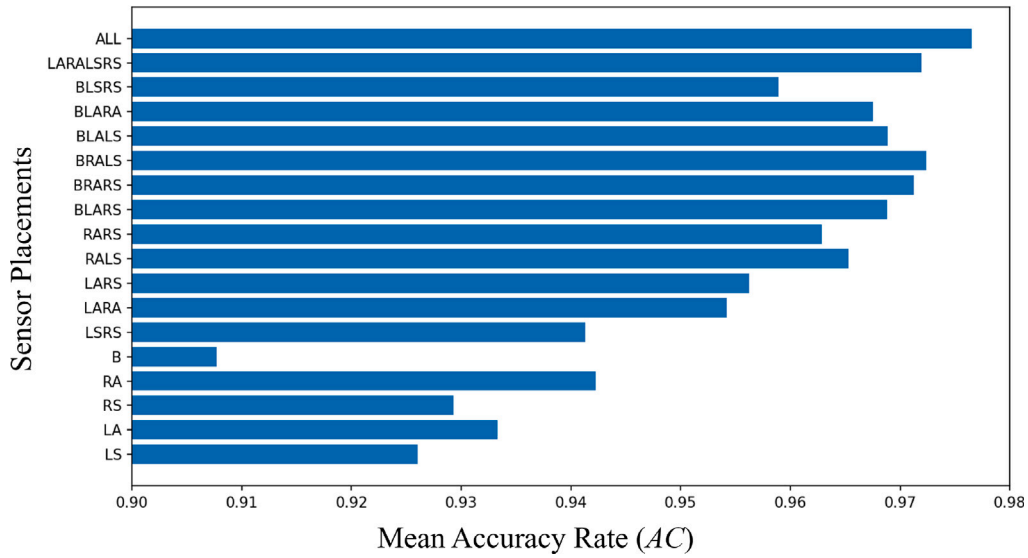


Fig. 6. AC results of RF models trained with different combinations of sensor placements.

Table 4

Sub-activities of UB and LB associated with main activities, including raw and segmented data counts.

Main activity	Sub-activity UB	Sub-activity LB	Total raw data	Total segmented data
A1	UB1	LB3	14,410	2684
A2	UB2	LB2	11,002	2005
A3	UB2	LB2	15,550	2930
A4	UB2	LB1	23,338	4494
A5	UB1	LB3	12,740	2386
A6	UB2	LB1	20,152	3854
A7	UB3	LB2	27,202	4922
A8	UB3	LB3	19,512	3672
Total			143,906	26,947

Table 5

Utilized classifiers and their respective hyperparameter for evaluating the performance of proposed HML-based network.

Classifier	Hyperparameter
KNN	MA: n_neighbours = 6; MU: n_neighbours = 7; ML: n_neighbours = 7
SVM	C = 5; Kernel = {'linear', 'poly', 'rbf'}
DT	MA: max_depth = 10; MU: max_depth = 7; ML: max_depth = 7; Criterion = 'entropy'
RF	n_estimators = 200; Criterion = 'entropy'
CNN-LSTM	Epoch = 100; Batch = 5; 3 1D-CNN layers [1024, 512, 1024]; 3 LSTM layers [1024, 512, 25]; Dense layer [ReLU, 64]; Dense layer [softmax, 8]
Multi-ResAtt	Epoch = 100; Batch = 5; 5 parallel residual modules; 1D average pooling; bi-GRU + attention layer; Dense layer [ReLU, 64]; Dense layer [softmax, 8]

Table 6

Performance metrics of the MA model using RF classifier (MA-RL) trained with IMU, sEMG, and combined IMU + sEMG features.

Sensor	RC	PR	F1	AC
IMU	90.38	91.59	90.98	97.66
sEMG	54.99	56.81	55.89	83.72
sEMG + IMU	91.32	92.30	91.81	97.86

$$RC = (1/N) \sum_{i=1}^N TP_i / (TP_i + FN_i) \quad (13)$$

$$PR = (1/N) \sum_{i=1}^N TP_i / (TP_i + FP_i) \quad (14)$$

$$F1 = (2 \times RC \times PR) / (RC + PR) \quad (15)$$

where N denotes the number of classes (eight activities), and TP_i , TN_i , FP_i , FN_i denote the true positives, true negatives, false positives, and false negatives of class i , respectively. These metrics collectively provide a comprehensive assessment of the network's performance across various firefighting activities.

4.1. Impact of SN placement

To evaluate the influence of SN placement combinations, the RF model was trained with various configurations. Fig. 6 illustrates the results, showing that using SNs on the B alone resulted in the lowest accuracy (AC) of 90.77%. This outcome aligns with expectations due to the similarity in body movements across these activities. Conversely, utilizing SNs on the RA led to a higher AC of 94.22%, highlighting improved accuracy for right-handed individuals, as all firefighters in the study were right-handed. Meanwhile, using SNs on the B, RA, and LS (BRALS) achieved a high AC of 97.24%, while utilizing SNs on all five positions (ALL) achieved the highest AC of 97.86%. Nonetheless, the BRASL configuration, which uses only three SNs, proves to be a practical solution.

Furthermore, this study evaluates the efficacy of integrating sEMG features. Table 6 presents the classification performance metrics of the MA-RF classifier trained with IMU, sEMG, and combined IMU + sEMG features. The results indicate superior performance when both IMU and sEMG features are utilized. In contrast, the MA-RF classifier trained solely with sEMG features exhibits poorer overall performance. To

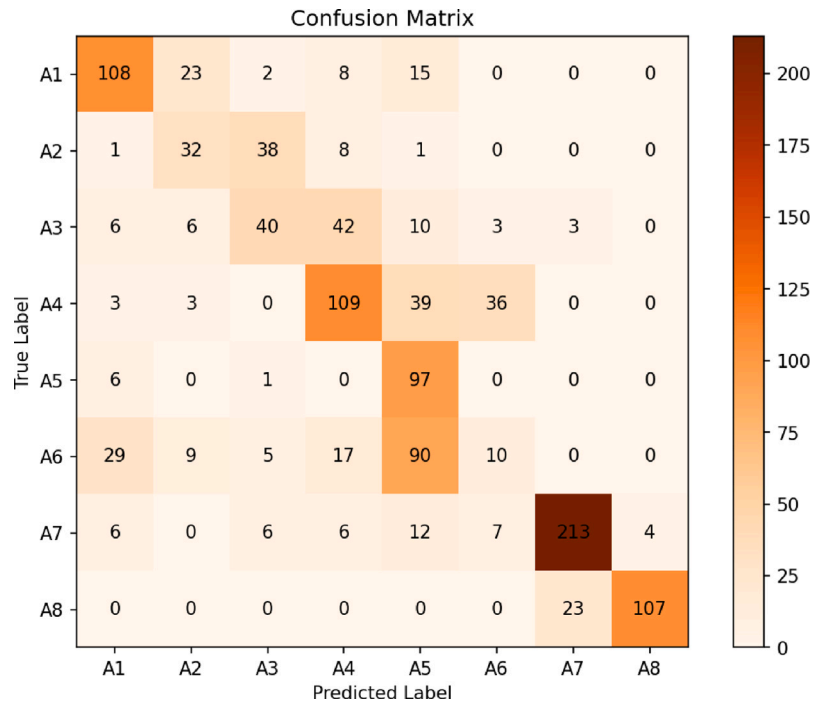


Fig. 7. Confusion matrix for the MA-RF model trained with sEMG features.

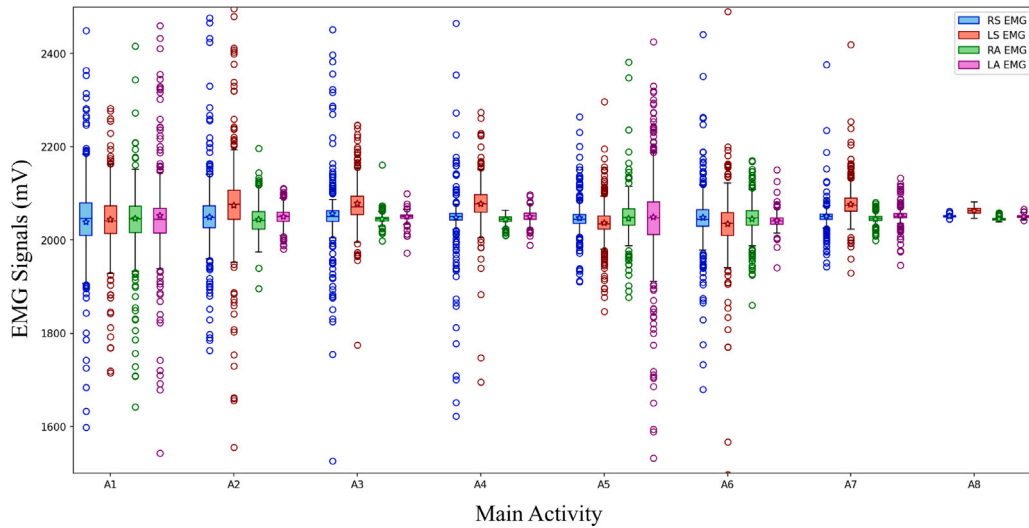


Fig. 8. Boxplot of the sEMG data STD at RS, LS, RA and LA across A1 to A8.

explore this further, Fig. 7 displays the confusion matrix highlighting varying accuracies across different activities. A5 ($AC = 93.2\%$), A7 ($AC = 83.9\%$), and A8 ($AC = 82.3\%$) show relatively higher accuracies compared to A1 ($AC = 69.2\%$), A2 ($AC = 40.0\%$), A3 ($AC = 36.3\%$), A4 ($AC = 57.4\%$) and A6 ($AC = 6.25\%$), highlighting the challenges with using only sEMG for diverse FAR tasks. Fig. 8 depicts the mean sEMG signals captured from RS, LS, RA, and LA across various activities. The analysis indicates a prominent consistency in sEMG patterns between LA and RA, especially shown in A2, A3, A4, and A6. This observation provides additional support for the earlier findings. A7 and A8, in contrast, exhibit simpler patterns, primarily influenced by features in the LB as prioritized by the trained models.

The results from the study highlight that the U2 sub-activity had a lower recognition rate (see Table 4). Comparison of A1 and A5, both involving sub-activities UB1 and LB3, revealed that A1 exhibited higher movement frequency but lower AC compared to A5. Interestingly,

analysis of all firefighters' sEMG signals indicated consistent patterns across different activities, likely due to their daily extensive weight training. This suggests that while sEMG features effectively detect activity presence, they are less reliable in distinguishing specific firefighting activities among firefighters.

4.2. Performance evaluation of MA, MU and ML models

The study evaluates the performance of MA, MU, and ML models to identify the optimal combination of classifiers for constructing the proposed HML-based network. Fig. 9 presents the results, showing that all classifiers achieve RC values exceeding 80%, except for MA-DT. The RF method for MU and ML exhibits the highest RC values at 95.47% and 92.98%, respectively. Although the MA-RF model demonstrates a slightly lower RC of 91.32% compared to the MA-SVM-RBF model ($RC = 91.83\%$), it outperforms the MA-SVM-RBF in other metrics, with

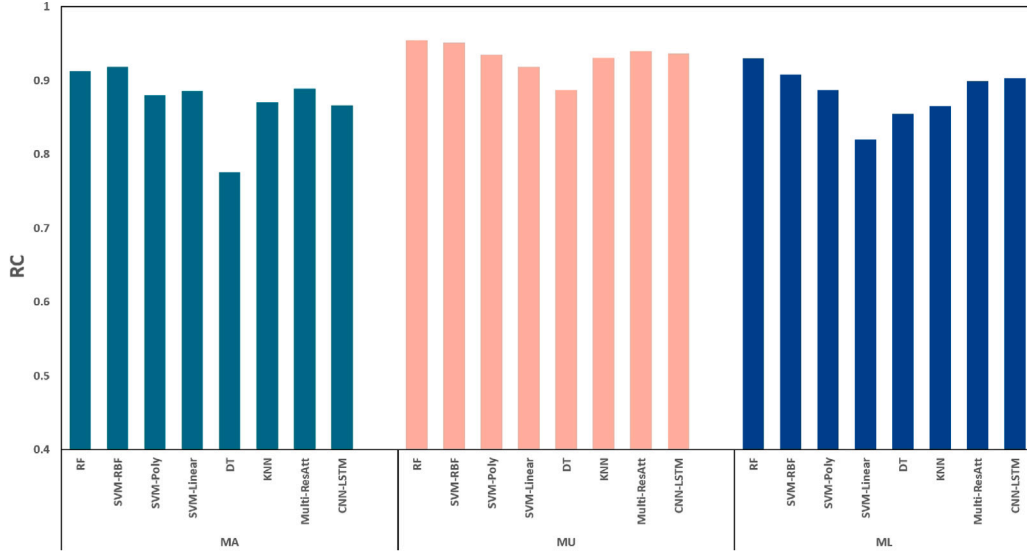


Fig. 9. RC values of MA, MU, and ML models trained using various classifiers.

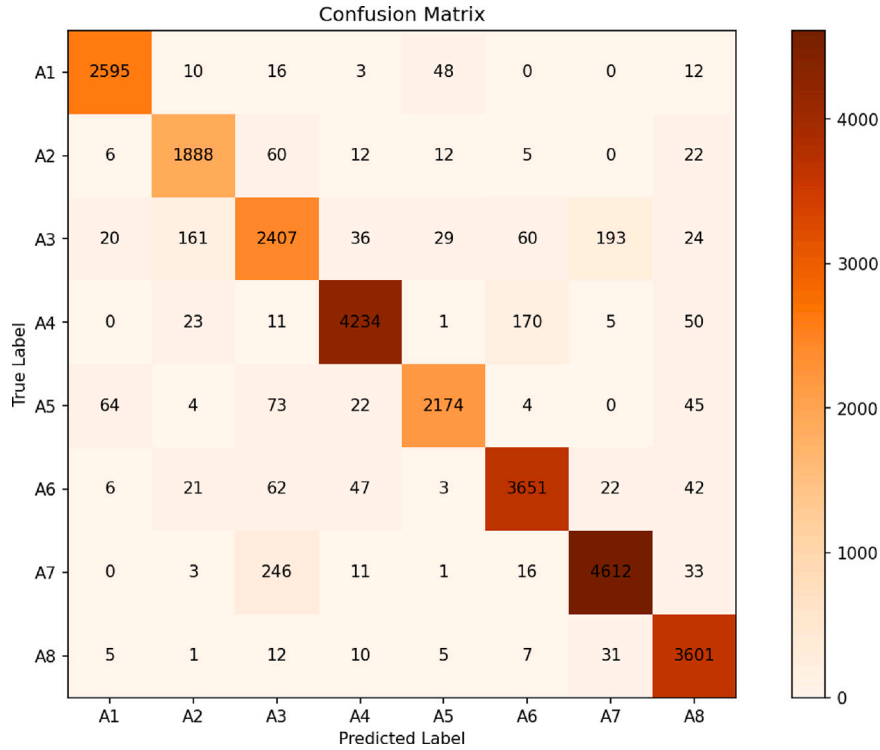


Fig. 10. Confusion matrix for the proposed HML-SVM-RBF1-RF2 network.

$PR = 92.3\%$, $F1 = 91.81\%$, and $AC = 97.86\%$ compared to $PR = 88.87\%$, $F1 = 90.33\%$, and $AC = 97.55\%$.

Analysis indicates that MU-RF and ML-RF models perform better in terms of RC metrics for classifying sub-activities compared to the MA-RF model, consistent with prior findings [47]. Meanwhile, DL models such as Multi-ResAtt and CNN-LSTM show lower performance despite their higher computational complexity and processing costs, aligning with observations from related research [8]. Based on these findings, RF and SVM-RBF classifiers are selected for constructing the HML-based network, with various configurations detailed in Table 7.

4.3. Performance evaluation of HML-based networks

Table 8 presents the performance evaluation of HML-based networks alongside comparisons with related studies on recognizing complex activities. Among the configurations tested, the HML-SVM-RBF1-RF2 network demonstrates superior results with a AC of 98.29%, RC of 93.48%, and $F1$ of 92.19%, outperforming other configurations. Studies by [42,64] in FAR showed comparatively lower PR at 83.91% and 87.84%, respectively, when compared to our proposed network. In contrast, studies by [43,46] demonstrated marginally better PR but lower RC and $F1$ scores. Meanwhile, the study by [49] achieved a high

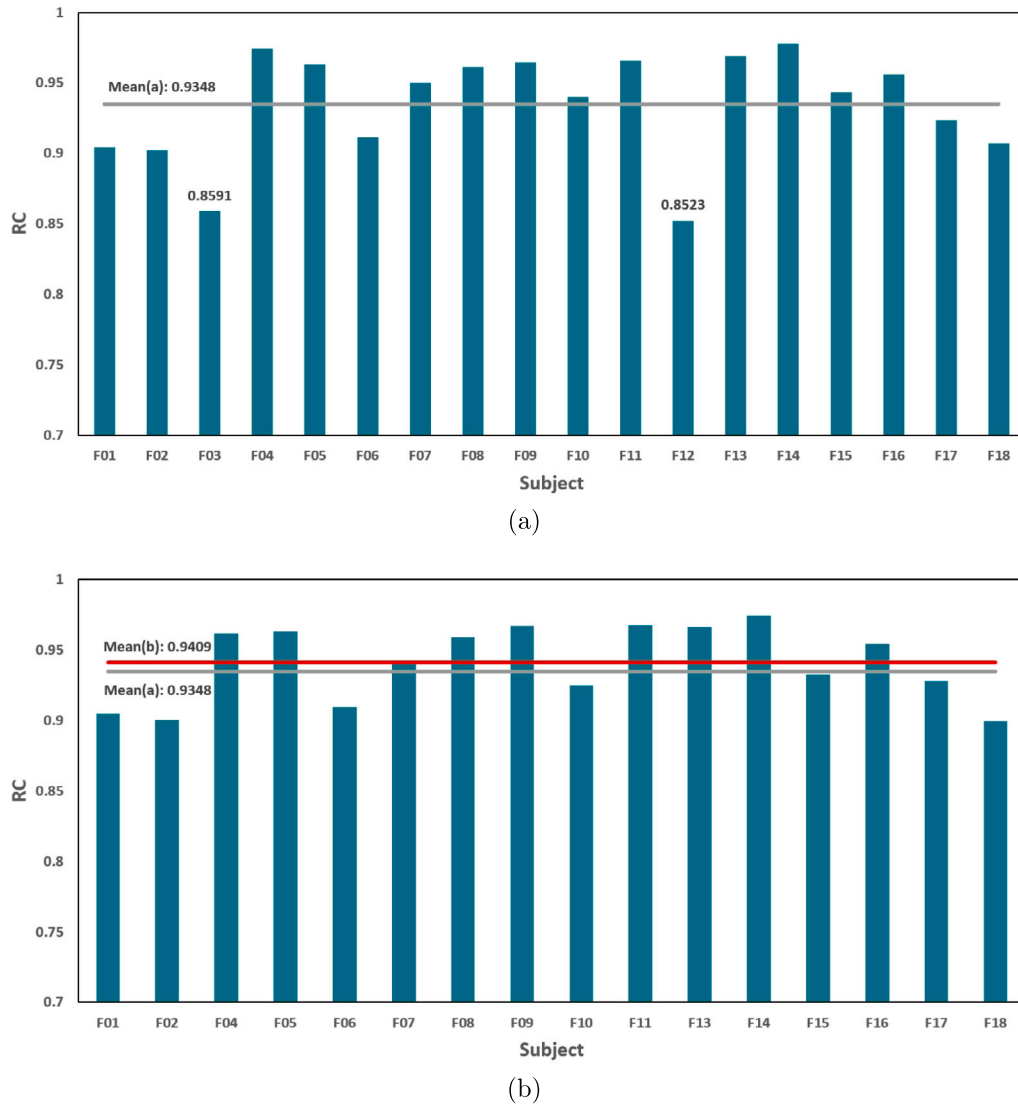


Fig. 11. Performance of the HML-SVM-RBF1-RF2 network in RC using leave-one-out cross-validation: (a) for all 18 subjects, and (b) for 16 subjects excluding F03 and F12.

Table 7

Proposed HML-based network configuration for MA, MU and ML models.

HML-based network	MA	MU	ML
HML-RF3	RF	RF	RF
HML-SVM-RBF3	SVM-RBF	SVM-RBF	SVM-RBF
HML-SVM-RBF1-RF2	SVM-RBF	RF	RF

PR of 97.7%, yet their dataset was limited to only two participants, affecting the model generalizability. Similarly, the study by [50] achieved acceptable results for classifying sixteen construction activities but with less relevance to firefighting tasks. Overall, our HML-SVM-RBF1-RF2 network shows robust performance in FAR compared to other studies.

Fig. 10 depicts the confusion matrix for the HML-SVM-RBF1-RF2 network, illustrating its effective performance in FAR. Specifically, A8 achieves a high AC with minimal misclassification rate (1.93%), indicating the network's capability in identifying abnormal stationary behaviors. Conversely, distinguishing A3 poses challenges due to its similarity with A2 and A7, resulting in the lowest overall AC. This highlights the necessity of utilizing individual models for MU and ML to increase FAR AC. In addition, an interesting observation reveals a 5.33% misclassification rate (246 out of 4612) of A7 as A3, attributed to

firefighters' common practice of placing nozzles on shoulders, affecting forearm muscle strength distinct from other UB2.

4.4. Discussion on activity behavior diversity

HAR studies typically enforce standardized activity performance protocols to ensure consistent recognition accuracy. However, real-world scenarios often reveal substantial variations in participant behavior, impacting recognition outcomes. In this study, firefighters occasionally exhibited signs of fatigue during training, leading to reduced FAR accuracy for specific firefighters. To address this variability, a detailed analysis was conducted using the leave-one-out cross-validation method with the HML-SVM-RBF1-RF2 network, as detailed in Fig. 11(a) and Table 9.

The analysis highlighted that F03 and F12 demonstrated lower RC values compared to others, with the average RC across 18 subjects being 93.48%. Fig. 12 presents the confusion matrix for F12, revealing challenges in accurately predicting A3 and A5, which were often misclassified as A2 and A3, respectively. Subsequent video analysis compared the execution of A2, A3, and A5 by both F12 (with the lowest RC) and F14 (with the highest RC). Fig. 13(a) depicts the incorrect rescuing behavior of F12, who pulled the dummy with one hand, while Fig. 13(b) shows the correct rescuing method. Additionally, Fig. 13(c)

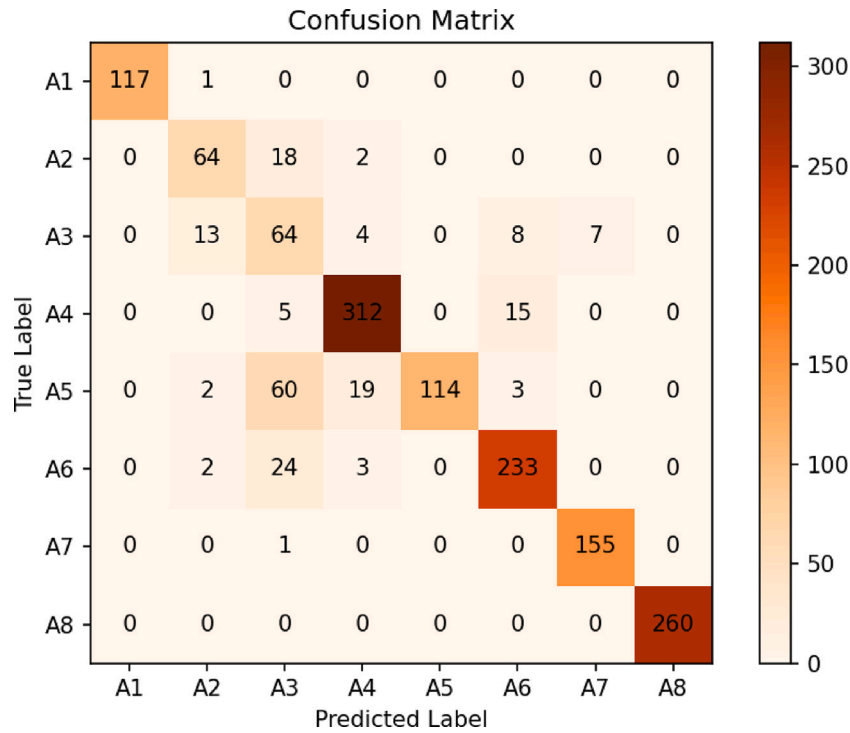


Fig. 12. Confusion matrix for F12 using the proposed HML-SVM-RBF1-RF2 network.

Table 8

Performance comparison of the proposed HML-based network with existing studies on complex HAR across different applications.

Study	Application	Activity (Subjects)	Sensing method	RC	PR	F1	AC
Lasek Jan & Gagolewski (2015) [64]	Firefighting	19 FAs (10)	7 6-DOF IMUs at legs, hands, arms, and back.	/	83.91	/	/
Geng et al. (2016) [42]	Firefighting	7 ADLs (N/A)	4 RF transmitters at chest, forehead, right wrist, and right ankle.	/	87.84	/	/
Qi et al. (2019) [46]	Gym setting	19 PAs (10)	ACC at wrist and ACC+ECG at chest.	89.75	91.69	90.78	/
Asuroglu (2022) [43]	Assisted setting	24 ADLs (52)	ACC at wrist.	91	90.9	91	91.3
Mastakouris et al. (2023) [49]	Assembly work	5 CAs (2)	Wrist worn smartphone.	/	97.7	/	/
				72.27	72.52	63.29	76.37
Mekruksavanich et al. (2023) [50]	Construction work	16 CAs (13)	3 9-DOF IMUs at hip, upper arm, and the rear of the shoulder.	/	/	88.67	88.64
HML-RF3	Firefighting	8 FAs (18)	4 6-DOF IMUs+sEMG at forearms and shanks;	92.93	91.15	92.03	97.91
HML-SVM-RBF3			1 6-DOF IMU at back.	92.58	89.32	90.92	97.68
HML-SVM-RBF1-RF2*				93.48	90.94	92.19	98.29

Note: N/A: not available; FAs: firefighting activities; CAs: complex activities; PAs: physical activities; ACC: accelerometer; RF: radio frequency; ECG: electrocardiogram.

illustrates F12 incorrectly dragging the hose backward, contrary to the correct method demonstrated by F14 in Fig. 13(d), where the hose was placed on the shoulder and moved forward.

In scenario A5, F12 mistakenly pulled the hose by moving his entire body backward (see Fig. 13(e)), instead of relying solely on upper body strength to draw the hose closer to the window while maintaining his position (see Fig. 13(f)). This incorrect method was noticeably similar to the A3, which deviated from the proper training standards, as confirmed by the firefighting instructors. Further analysis of firefighter performance, excluding F03 and F12, is illustrated in Fig. 11(b). This analysis shows that most firefighters achieved over 90% in RC, with the mean RC value increasing by 0.61%. These findings underscore the impact of behavioral diversity on firefighting activity recognition.

Conversely, the misclassification of FAR presents an excellent opportunity to assess firefighter performance during training. This allows trainers and trainees to conduct thorough evaluations and reflections, facilitating detailed performance analysis. These results suggest that diverse behaviors pose challenges in accurately identifying human activities characterized by intricate movements, particularly tasks involving diverse and specific gestures, such as those performed by firefighters in real-world scenarios.

5. Conclusion

This study introduces an IoT-FAR system and an HML-based network for recognizing SCBA endurance training activities in firefighting, leveraging multi-modal sensor fusion approach. The study utilizes the

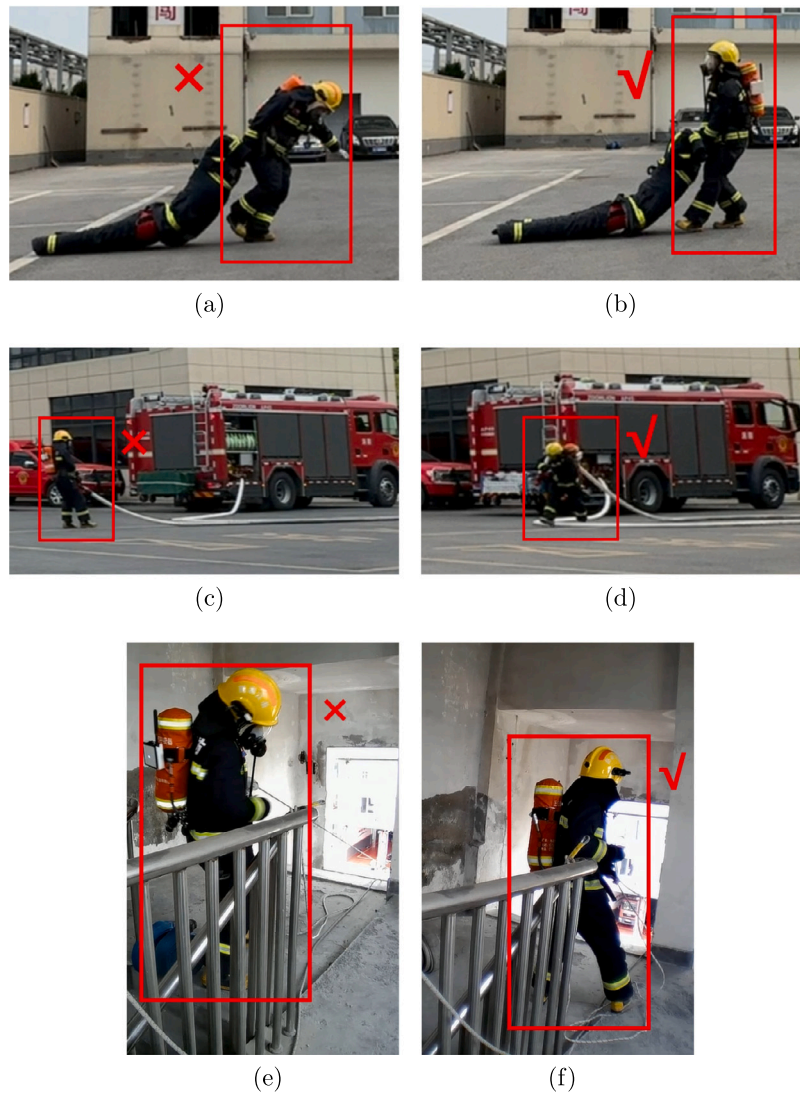


Fig. 13. Performance evaluation of various activities, including: (a) incorrect execution of A2 by F12, (b) correct execution of A2 by F14, (c) incorrect execution of A3 by F12, (d) correct execution of A3 by F14, (e) incorrect execution of A5 by F12, and (f) correct execution of A5 by F14.

Table 9
Performance of HML-SVM-RBF1-RF2 network using leave-one-out cross validation.

Firefighter	RC	PR	F1	AC
F01	90.47	85.74	88.04	97.90
F02	90.24	82.76	86.34	96.79
F03	85.91	83.85	84.87	96.68
F04	97.41	96.08	96.74	99.18
F05	96.30	95.38	95.84	98.99
F06	91.16	88.97	90.02	97.31
F07	95.04	91.54	93.26	98.39
F08	96.13	89.15	92.51	98.94
F09	96.48	95.60	96.04	99.02
F10	93.99	94.38	93.96	98.57
F11	96.58	88.03	94.18	99.03
F12	85.23	86.58	85.90	96.58
F13	96.90	95.62	96.26	99.04
F14	97.78	97.02	97.40	99.45
F15	94.35	93.26	93.80	98.53
F16	95.61	94.86	95.23	99.08
F17	92.37	87.26	89.74	98.18
F18	90.72	90.76	90.74	97.64
Mean	93.48	90.94	92.27	98.29

SVM-RBF model in the MA and RF model in both MU and ML models to establish the HML-SVM-RBF1-RF2 network, achieving a highest accuracy of 98.29%. Integrating multiple sensors increases FAR rates but adds computational complexity. Utilizing sensors on the body and dominant hand proves effective for FAR, despite sEMG features lacking distinct patterns across different firefighters, necessitating further research for improved differentiation in real-world FAR. The study also highlights the potential use of misclassification results as an assessment tool for training activities.

Nevertheless, this study is limited by its focus on SCBA endurance training, which were conducted in controlled fire scenarios without actual fire and smoke. Consequently, the HR data patterns for firefighters in simulated firefighting environments are less varied and realistic. Additionally, the sensors and hardware of the proposed IoT-FAR system require refinement in heat resistance, waterproofing, and explosion-proofing to address practical challenges in real firefighting scenarios. While the proposed HML model proves effective based on the multi-modal sensor fusion approach using IMU and sEMG signals, there is no public dataset with a similar sensing approach targeting firefighting activities, to the authors' knowledge.

To validate the generalization performance of the proposed HML method, future research aims to expand datasets by integrating a broader range of training activities, especially in more authentic and

realistic fire simulations involving real smoke and fire. Further work will focus on refining the proposed IoT-FAR system to integrate a large-scale firefighting network architecture, potentially extending across regions or even nationally, forming a comprehensive smart IoT infrastructure for cloud-based nationwide smart firefighting platform. Additional research will explore the design of sensory modules on smart firefighting equipment to facilitate the evolution towards the Internet of Everything (IoE). Another interesting opportunity for future work is developing an advanced recognition system capable of distinguishing correct and incorrect behaviors to assess training performance.

CRedit authorship contribution statement

Xiaoqing Chai: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Boon Giin Lee:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition. **Chenhang Hu:** Software, Methodology, Formal analysis, Data curation. **Matthew Pike:** Writing – review & editing, Validation, Supervision. **David Chieng:** Writing – review & editing, Supervision. **Renjie Wu:** Visualization, Validation, Software, Data curation. **Wan-Young Chung:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

I have nothing to declare.

Data availability

The SFTAA dataset is available upon request through our GitHub repository at <https://github.com/HCI-Laboratory/Smart-Firefighting/tree/main/Firefighter-Activity-Recognition>.

Acknowledgments

This work was supported by the University of Nottingham Ningbo China through Li Dak Sum Innovation Fellowship under Grant LDS20 2307. The authors express deep gratitude to the Huzhou Fire Brigade for their invaluable support and contributions to this study. They also acknowledge each firefighter involved in the experiments for providing firsthand insights and unique perspectives derived from their experience. These contributions not only improved the data quality but also added a practical dimension to the theoretical framework.

References

- [1] China Fire Protection Yearbook 2020, Ministry of Emergency Management Publishing House, 2020.
- [2] M. Fan, Q. Yang, S. Feng, C. Zhao, J. Pu, Research on causalities of firefighters in various firefighting and rescue tasks, *Ind. Saf. Environ. Protect.* 41 (2) (2015).
- [3] W. Fu, Influencing factors and coping strategies of firefighter training lassitude, *J. Sci. Technol.* 7 (2022).
- [4] H. Yuan, Influencing factors and coping strategies of firefighter training lassitude, *J. China People's Police Univ.* 30 (2) (2014).
- [5] D. Chen, S. Yongchareon, E.M.-K. Lai, J. Yu, Q.Z. Sheng, Y. Li, Transformer with bidirectional GRU for nonintrusive, sensor-based activity recognition in a multiresident environment, *IEEE Internet Things J.* 9 (23) (2022) 23716–23727, <http://dx.doi.org/10.1109/JIOT.2022.3190307>.
- [6] S.M.M. Islam, K.H. Talukder, Exploratory analysis of smartphone sensor data for human activity recognition, *IEEE Access* 11 (2023) 99481–99498, <http://dx.doi.org/10.1109/ACCESS.2023.3314651>.
- [7] J. Huang, S. Lin, N. Wang, G. Dai, Y. Xie, J. Zhou, TSE-CNN: A two-stage end-to-end CNN for human activity recognition, *IEEE J. Biomed. Health Inf.* 24 (1) (2020) 292–299, <http://dx.doi.org/10.1109/JBHI.2019.2909688>.
- [8] R. Ge, Xgboost-based human activity recognition algorithm using wearable smart devices, in: *Proceedings of the 4th International Conference on Computing and Data Science (CONF-CDS 2022)*, Vol. 2, 2023, pp. 352–358, <http://dx.doi.org/10.54254/2755-2721/2/20220514>.
- [9] J. Zhu, H. Chen, W. Ye, A hybrid CNN-LSTM network for the classification of human activities based on micro-Doppler radar, *IEEE Access* 8 (2020) 24713–24720, <http://dx.doi.org/10.1109/ACCESS.2020.2971064>.
- [10] P. Turaga, R. Chellappa, V.S. Subrahmanian, O. Udrea, Machine recognition of human activities: A survey, *IEEE Trans. Circuits Syst. Video Technol.* 18 (11) (2008) 1473–1488, <http://dx.doi.org/10.1109/TCST.2008.2005594>.
- [11] M. Abdel-Basset, H. Hawash, V. Chang, R.K. Chakraborty, M. Ryan, Deep learning for heterogeneous human activity recognition in complex IoT applications, *IEEE Internet Things J.* 9 (8) (2022) 5653–5665, <http://dx.doi.org/10.1109/JIOT.2020.3038416>.
- [12] T. Gu, L. Wang, Z. Wu, X. Tao, J. Lu, A pattern mining approach to sensor-based human activity recognition, *IEEE Trans. Knowl. Data Eng.* 23 (9) (2011) 1359–1372, <http://dx.doi.org/10.1109/TKDE.2010.184>.
- [13] R. Huan, C. Jiang, L. Ge, J. Shu, Z. Zhan, P. Chen, K. Chi, R. Liang, Human complex activity recognition with sensor data using multiple features, *IEEE Sens. J.* 22 (1) (2022) 757–775, <http://dx.doi.org/10.1109/JSEN.2021.3130913>.
- [14] R. Nair, M. Ragab, O. Mujallid, K. Mohammad, R. Mansour, G. Viju, Impact of wireless sensor data mining with hybrid deep learning for human activity recognition, *Wirel. Commun. Mob. Comput.* 2022 (2022) <http://dx.doi.org/10.1155/2022/9457536>.
- [15] R. Zhu, Z. Xiao, Y. Li, M. Yang, Y. Tan, L. Zhou, S. Lin, H. Wen, Efficient human activity recognition solving the confusing activities via deep ensemble learning, *IEEE Access* 7 (2019) 75490–75499, <http://dx.doi.org/10.1109/ACCESS.2019.2922104>.
- [16] Y. Li, L. Yu, J. Liao, G. Su, H. Ammarah, L. Liu, S. Wang, A single smartwatch-based segmentation approach in human activity recognition, *Pervasive Mob. Comput.* 83 (2022) 101600, <http://dx.doi.org/10.1016/j.pmcj.2022.101600>.
- [17] R. Gravina, P. Alinia, H. Ghasemzadeh, G. Fortino, Multi-sensor fusion in body sensor networks: State-of-the-art and research challenges, *Inf. Fusion* 35 (2017) 68–80, <http://dx.doi.org/10.1016/j.inffus.2016.09.005>, URL <https://www.sciencedirect.com/science/article/pii/S156625351630077X>.
- [18] S. Qiu, H. Zhao, N. Jiang, Z. Wang, L. Liu, Y. An, H. Zhao, X. Miao, R. Liu, G. Fortino, Multi-sensor information fusion based on machine learning for real applications in human activity recognition: State-of-the-art and research challenges, *Inf. Fusion* 80 (2022) 241–265, <http://dx.doi.org/10.1016/j.inffus.2021.11.006>, URL <https://www.sciencedirect.com/science/article/pii/S1566253521002311>.
- [19] P. Bharti, D. De, S. Chellappan, S.K. Das, Human: Complex activity recognition with multi-modal multi-positional body sensing, *IEEE Trans. Mob. Comput.* 18 (4) (2019) 857–870, <http://dx.doi.org/10.1109/TMC.2018.2841905>.
- [20] C. Zhu, W. Sheng, Realtime recognition of complex human daily activities using human motion and location data, *IEEE Trans. Biomed. Eng.* 59 (9) (2012) 2422–2430, <http://dx.doi.org/10.1109/TBME.2012.2190602>.
- [21] S. Cheng, E. Bolívar-Nieto, R.D. Gregg, Real-time activity recognition with instantaneous characteristic features of thigh kinematics, *IEEE Trans. Neural Syst. Rehabil. Eng.* 29 (2021) 1827–1837, <http://dx.doi.org/10.1109/TNSRE.2021.3107780>.
- [22] T.N. Do, R. Liu, C. Yuen, U.-X. Tan, Personal dead reckoning using IMU device at upper torso for walking and running, in: *2016 IEEE SENSORS*, 2016, pp. 1–3, <http://dx.doi.org/10.1109/ICSENS.2016.7808521>.
- [23] R. Liu, C. Yuen, T.-N. Do, U.-X. Tan, Fusing similarity-based sequence and dead reckoning for indoor positioning without training, *IEEE Sens. J.* 17 (13) (2017) 4197–4207, <http://dx.doi.org/10.1109/JSEN.2017.2706303>.
- [24] R. Liu, S.H. Marakkalage, M. Padmal, T. Shaganan, C. Yuen, Y.L. Guan, U.-X. Tan, Collaborative SLAM based on WiFi fingerprint similarity and motion information, *IEEE Internet Things J.* 7 (3) (2020) 1826–1840, <http://dx.doi.org/10.1109/JIOT.2019.2957293>.
- [25] R. Liu, C. Yuen, T.-N. Do, M. Zhang, Y.L. Guan, U.-X. Tan, Cooperative positioning for emergency responders using self IMU and peer-to-peer radios measurements, *Inf. Fusion* 56 (2020) 93–102, <http://dx.doi.org/10.1016/j.inffus.2019.10.009>, URL <https://www.sciencedirect.com/science/article/pii/S1566253518306158>.
- [26] S.S. Bangaru, C. Wang, F. Aghazadeh, Data quality and reliability assessment of wearable EMG and IMU sensor for construction activity recognition, *Sensors* 20 (18) (2020) <http://dx.doi.org/10.3390/s20185264>.
- [27] A. Vijayvargiya, P. Singh, R. Kumar, N. Dey, Hardware implementation for lower limb surface EMG measurement and analysis using explainable AI for activity recognition, *IEEE Trans. Instrum. Meas.* 71 (2022) 1–9, <http://dx.doi.org/10.1109/TIM.2022.3198443>.
- [28] O.W. Samuel, M.G. Asogbon, Y. Geng, A.H. Al-Timemy, S. Pirbhulal, N. Ji, S. Chen, P. Fang, G. Li, Intelligent EMG pattern recognition control method for upper-limb multifunctional prostheses: Advances, current challenges, and future prospects, *IEEE Access* 7 (2019) 10150–10165, <http://dx.doi.org/10.1109/ACCESS.2019.2891350>.
- [29] L. Pei, S. Xia, L. Chu, F. Xiao, Q. Wu, W. Yu, R. Qiu, MARS: Mixed virtual and real wearable sensors for human activity recognition with multidomain deep learning model, *IEEE Internet Things J.* 8 (11) (2021) 9383–9396, <http://dx.doi.org/10.1109/JIOT.2021.3055859>.
- [30] N. Mani, P. Haridoss, B. George, Smart suspenders with sensors and machine learning for human activity monitoring, *IEEE Sens. J.* 23 (9) (2023) 10159–10167, <http://dx.doi.org/10.1109/JSEN.2023.3263231>.
- [31] X. Yao, X. Shi, F. Zhou, Human activities classification based on complex-valued convolutional neural network, *IEEE Sens. J.* 20 (13) (2020) 7169–7180, <http://dx.doi.org/10.1109/JSEN.2020.2967054>.

- [32] W. Qi, H. Su, A. Aliverti, A smartphone-based adaptive recognition and real-time monitoring system for human activities, *IEEE Trans. Hum.-Mach. Syst.* 50 (5) (2020) 414–423, <http://dx.doi.org/10.1109/THMS.2020.2984181>.
- [33] J. Zhang, Y. Liu, H. Yuan, Attention-based residual BiLSTM networks for human activity recognition, *IEEE Access* 11 (2023) 94173–94187, <http://dx.doi.org/10.1109/ACCESS.2023.3310269>.
- [34] M.A.A. Al-qaness, A. Dahou, M.A. Elaziz, A.M. Helmi, Multi-ResAtt: Multilevel residual network with attention for human activity recognition using wearable sensors, *IEEE Trans. Ind. Inform.* 19 (1) (2023) 144–152, <http://dx.doi.org/10.1109/TII.2022.3165875>.
- [35] Y. Wang, X. Wang, H. Yang, Y. Geng, H. Yu, G. Zheng, L. Liao, MhaGNN: A novel framework for wearable sensor-based human activity recognition combining multi-head attention and graph neural networks, *IEEE Trans. Instrum. Meas.* 72 (2023) 1–14, <http://dx.doi.org/10.1109/TIM.2023.3276004>.
- [36] S. Mekruksavanich, A. Jitpattanakul, K. Sithithakerngkiet, P. Youplao, P. Yupapin, ResNet-SE: Channel attention-based deep residual network for complex activity recognition using wrist-worn wearable sensors, *IEEE Access* 10 (2022) 51142–51154, <http://dx.doi.org/10.1109/ACCESS.2022.3174124>.
- [37] J. Liang, L. Zhang, C. Han, C. Bu, H. Wu, A. Song, A collaborative compression scheme for fast activity recognition on mobile devices via global compression ratio decision, *IEEE Trans. Mob. Comput.* (2023) 1–14, <http://dx.doi.org/10.1109/TMC.2023.3271306>.
- [38] M.-K. Yi, W.-K. Lee, S.O. Hwang, A human activity recognition method based on lightweight feature extraction combined with pruned and quantized CNN for wearable device, *IEEE Trans. Consum. Electron.* 69 (3) (2023) 657–670, <http://dx.doi.org/10.1109/TCE.2023.3266506>.
- [39] E. Uzunhisarcikli, E. Kavuncuoğlu, A.T. Özdemir, Investigating classification performance of hybrid deep learning and machine learning architectures on activity recognition, *Comput. Intell.* 38 (4) (2022) 1402–1449, <http://dx.doi.org/10.1111/coin.12517>.
- [40] B. Vidyaa, S. P., Wearable multi-sensor data fusion approach for human activity recognition using machine learning algorithms, *Sensors Actuators A* 341 (2022) 113557, <http://dx.doi.org/10.1016/j.sna.2022.113557>.
- [41] Q. Mascaret, M. Biemann, C.-L. Fall, L.J. Bouyer, B. Gosselin, Real-time human physical activity recognition with low latency prediction feedback using raw IMU data, in: 2018 40th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, EMBC, 2018, pp. 239–242, <http://dx.doi.org/10.1109/EMBC.2018.8512252>.
- [42] Y. Geng, J. Chen, R. Fu, G. Bao, K. Pahlavan, Enlighten wearable physiological monitoring systems: On-body RF characteristics based human motion classification using a support vector machine, *IEEE Trans. Mob. Comput.* 15 (3) (2016) 656–671, <http://dx.doi.org/10.1109/TMC.2015.2416186>.
- [43] T. Aşuroğlu, Complex human activity recognition using a local weighted approach, *IEEE Access* 10 (2022) 101207–101219, <http://dx.doi.org/10.1109/ACCESS.2022.3208686>.
- [44] Y. Dong, R. Zhou, C. Zhu, L. Cao, X. Li, Hierarchical activity recognition based on belief functions theory in body sensor networks, *IEEE Sens. J.* 22 (15) (2022) 15211–15221, <http://dx.doi.org/10.1109/JSEN.2022.3186086>.
- [45] F. Samie, L. Bauer, J. Henkel, Hierarchical classification for constrained IoT devices: A case study on human activity recognition, *IEEE Internet Things J.* 7 (9) (2020) 8287–8295, <http://dx.doi.org/10.1109/JIOT.2020.2989053>.
- [46] J. Qi, P. Yang, M. Hanneghan, S. Tang, B. Zhou, A hybrid hierarchical framework for gym physical activity recognition and measurement using wearable sensors, *IEEE Internet Things J.* 6 (2) (2019) 1384–1393, <http://dx.doi.org/10.1109/JIOT.2018.2846359>.
- [47] J.V. Jeyakumar, A. Sarker, L.A. Garcia, M. Srivastava, X-CHAR: A concept-based explainable complex human activity recognition model, *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol.* 7 (2023) <http://dx.doi.org/10.1145/3580804>.
- [48] V.S. Nguyen, H. Kim, D. Suh, Attention mechanism-based bidirectional long short-term memory for cycling activity recognition using smartphones, *IEEE Access* 11 (2023) 136206–136218, <http://dx.doi.org/10.1109/ACCESS.2023.3338137>.
- [49] A. Mastakouris, G. Andriosopoulou, D. Masouros, P. Benardos, G.-C. Vosniakos, D. Soudris, Human worker activity recognition in a production floor environment through deep learning, *J. Manuf. Syst.* 71 (2023) 115–130, <http://dx.doi.org/10.1016/j.jmsy.2023.08.020>.
- [50] S. Mekruksavanich, P. Jantawong, N. Hnoohom, A. Jitpattanakul, Wearable-based activity recognition of construction workers using lstm neural networks, in: 2022 37th International Technical Conference on Circuits/Systems, Computers and Communications (ITC-CSCC), 2022, pp. 1–4, <http://dx.doi.org/10.1109/ITC-CSCC55581.2022.9894868>.
- [51] M. Meina, A. Janusz, K. Rykaczewski, D. Slezak, B. Celmer, A. Krasuski, Tagging firefighter activities at the emergency scene: Summary of aaia'15 data mining competition at knowledge pit, in: 2015 Federated Conference on Computer Science and Information Systems (FedCSIS), 2015, pp. 367–373, <http://dx.doi.org/10.15439/2015F426>.
- [52] X. Chai, B.-G. Lee, M. Pike, R. Wu, D. Chieng, W.-Y. Chung, Pre-impact firefighter fall detection using machine learning on the edge, *IEEE Sens. J.* 23 (13) (2023) 14997–15009, <http://dx.doi.org/10.1109/JSEN.2023.3279858>.
- [53] Polar verity sense, 2023, [Online], Available: <https://www.polar.com/en/products/accessories/polar-verity-sense/>. (Accessed 21 April 2023).
- [54] Mpu 6050, 2023, [Online], Available: <https://invensense.tdk.com/products/motion-tracking/6-axis/mpu-6050/>. (Accessed 21 April 2023).
- [55] J. Song, A. Zhu, Y. Tu, H. Huang, M.A. Arif, Z. Shen, X. Zhang, G. Cao, Effects of different feature parameters of sEMG on human motion pattern recognition using multilayer perceptrons and LSTM neural networks, *Appl. Sci.* 10 (10) (2020) <http://dx.doi.org/10.3390/app10103358>.
- [56] X. Chai, R. Wu, M. Pike, H. Jin, W.-Y. Chung, B.-G. Lee, Smart wearables with sensor fusion for fall detection in firefighting, *Sensors* 21 (20) (2021) <http://dx.doi.org/10.3390/s2106770>.
- [57] Raspberry pi 4b, 2023, [Online], Available: <https://www.raspberrypi.com/products/raspberry-pi-4-model-b/>. (Accessed 10 May 2023).
- [58] Tinypico, 2023, [Online], Available: <https://www.tinypico.com/>. (Accessed 29 April 2023).
- [59] Bme688, 2023, [Online], Available: <https://www.adafruit.com/product/5046>. (Accessed 30 April 2023).
- [60] M.B. Rasheed, N. Javaid, T.A. Alghamdi, S. Mukhtar, U. Qasim, Z.A. Khan, M.H.B. Raja, Evaluation of human activity recognition and fall detection using android phone, in: 2015 IEEE 29th International Conference on Advanced Information Networking and Applications, 2015, pp. 163–170, <http://dx.doi.org/10.1109/AINA.2015.181>.
- [61] O. Dehzangi, V. Sahu, IMU-based robust human activity recognition using feature analysis, extraction, and reduction, in: 2018 24th International Conference on Pattern Recognition, ICPR, 2018, pp. 1402–1407, <http://dx.doi.org/10.1109/ICPR.2018.8546311>.
- [62] X. Chai, B.-G. Lee, M. Pike, R. Wu, X. Wu, A novel attitude feature extraction method for multi-IMU based fall detection system, in: 2023 IEEE 3rd International Conference on Power, Electronics and Computer Applications, ICPECA, 2023, pp. 99–103, <http://dx.doi.org/10.1109/ICPECA56706.2023.10075997>.
- [63] S.M. Lundberg, S.-I. Lee, A unified approach to interpreting model predictions, in: *Proceedings of the 31st International Conference on Neural Information Processing Systems*, Curran Associates Inc., 2017, pp. 4768–4777.
- [64] J. Lasek, M. Gagolewski, The winning solution to the AAIA'15 data mining competition: Tagging firefighter activities at a fire scene, in: 2015 Federated Conference on Computer Science and Information Systems (FedCSIS), 2015, pp. 375–380, <http://dx.doi.org/10.15439/2015F418>.